

W006 — Interactive UI Plan (Design ↔ As-Drawn Cross-Map Viewer)

Wave: W006

Status: Option A (cross-map engine) **delivered**; Option B (interactive UI) **scaffolded** — this document is the full plan to finish it.

Owner: MINERVA QCELL/RFCELL P&ID workstream

Honesty mandate: every UI affordance must show provenance and confidence, never imply a match is engineering-confirmed when it is heuristic.

1. Why this exists

W005 established that the **design** register (circuit-sequential tags such as `CV001` , `TT100`) and the **as-drawn** register (SVG-instance tags such as `CV560` , `TT514`) are **orthogonal numbering systems** with **0 % exact tag overlap**. W006 Option A turned that finding into a functional, confidence-scored bidirectional cross-map (`build_w006_crossmap.py`):

- 97 design tags · 141 as-drawn real tags
- 43 mapped (44.3 %): 0 HIGH · 39 MEDIUM (0.75) · 4 LOW (0.45)
- 54 unmapped design · 98 unclaimed as-drawn

These results are inherently **review-oriented**: a human engineer must triage MEDIUM/LOW pairs and confirm/reject them. A static JSON/Markdown report is not an efficient triage surface. The interactive viewer is the triage surface.

2. What the scaffold already delivers (`build_viewer.py` → `publish/interactive_viewer.html`)

Feature	State	Notes
Embedded annotated atlas SVG (QCELL)	✓ working	reuses <code>output_v6/QCELL/*_13layers.svg</code>
21-layer toggle checkboxes	✓ wired	drives <code>lyr-NN</code> CSS class visibility
Pan + zoom	✓ working	wheel = zoom, drag = pan (transform on <code>#svgpan</code>)
Search by tag (design OR as-drawn)	✓ working	highlights + scrolls to the matching row
Confidence-coloured cross-map table	✓ working	HIGH/MEDIUM/LOW/un-mapped pills
Metadata popup (reasons, tier, confidence)	✓ working	placeholder; per-pair <code>reasons[]</code> shown
Self-contained single HTML file	✓	no external JS/CSS deps

Explicitly NOT done yet (this is a scaffold): SVG element hit-test/highlight, side-by-side comparison overlay, image/PDF export, deep-link/URL state, validation write-back.

3. Full feature plan (phased)

Phase UI-1 — Tag ↔ SVG element linking (highest value)

- Annotate each drawable instrument bubble in the atlas SVG with `data-tag="CV560"` during `build_atlas_v6` (the catalog already carries `x/y/sheet` per instance).
- On row click / search hit: highlight the SVG element (outline + flash) and center it in the viewport.
- On SVG element click: select the corresponding table row and open the metadata popup.
- **Acceptance:** clicking `TT100` in the table outlines `TT514` 's bubble on the drawing within one viewport pan.

Phase UI-2 — Confidence triage workflow

- Add per-row **Confirm / Reject / Defer** controls that write to a local triage file (`data/crossmap/triage_decisions.json` , gitignored) — no mutation of the heuristic crossmap itself.
- Confirmed pairs become **KNOWN_SEEDS** fed back into `build_w006_crossmap.py` on the next run, which legitimately promotes corroborated pairs to HIGH (the engine already supports a `KNOWN_SEEDS` dict, currently empty by design).

- **Acceptance:** confirming a MEDIUM pair and re-running `make.sh` promotes it to HIGH with reason `KNOWN_SEED`.

Phase UI-3 — Comparison & filtering

- Filter chips: tier (HIGH/MEDIUM/LOW/unmapped), type (CV/TT/PT/...), circuit band (40K/4.5K/2K/WATER/VACUUM/ROOM), sheet (QCELL/RFCELL).
- “Unmapped only” and “needs review” quick views.
- Optional side-by-side: design schematic vs as-drawn, synced highlight.

Phase UI-4 — Export & sharing

- Export current view to PNG (client-side canvas) and the filtered table to CSV.
- Deep-link state in URL hash (`#tag=CV001&tier=MEDIUM`) so a reviewer can share an exact view.

Phase UI-5 — RFCELL + multi-sheet

- Currently embeds QCELL only. Add a sheet selector and load RFCELL atlas SVG.

4. Architecture decisions

- **Stdlib-first / no build step.** The viewer is generated by a Python module (`build_viewer.py`) and emitted as a single static HTML file. No npm, no bundler — consistent with the repo’s `./make.sh` reproducibility discipline.
- **Data flows one way for heuristics.** The viewer never edits the heuristic crossmap. Human decisions land in a separate triage file and only influence future runs via `KNOWN_SEEDS`. This preserves the honesty invariant enforced by `test_no_fabricated_high_without_corroboration`.
- **Provenance always visible.** Every mapped pair shows its `reasons[]` (e.g. `TYPE_MATCH; CIRCUIT_MATCH(40K); ORDER_MATCH(#0/3)`) so a reviewer sees why the heuristic proposed it.

5. Testing plan (standalone runners, no pytest)

- `tests/test_w006_viewer.py` (future): assert the generated HTML contains one table row per crossmap entry, that every `data-tag` resolves to a catalog instance, and that tier pill colours match the confidence tier. Guard with `_data_available()` like the other suites.
- Golden-file check: snapshot the row count + mapped count and assert against `reports/w006_crossmap_statistics.json` (cross-pollinates the golden-file pattern from the test suite).

6. Definition of done (for the full UI, beyond scaffold)

1. Tag ↔ SVG element linking works both directions (UI-1).
2. Triage decisions persist and feed `KNOWN_SEEDS` (UI-2).
3. Filters + unmapped view (UI-3).
4. PNG/CSV export + deep links (UI-4).
5. RFCELL sheet support (UI-5).
6. `tests/test_w006_viewer.py` green; viewer step in `make.sh` exits 0.